

Normal maps on L^∞ induced by operator-valued measures

A full answer to Problem 2 of arXiv:1411.4559

Automated mathematics run `fa_banach.001`, lane 13

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Status. Claimed full solution to the literal Problem 2, likely valid subject to human review. An inducing ultraweakly continuous linear map exists if and only if the operator-valued measure vanishes on all μ -null sets. Without this condition the answer is negative, even for a probability operator-valued measure on a one-dimensional Hilbert space.

1 Source problem

The source is D. Han, D. R. Larson, B. Liu, and R. Liu, *Dilations of frames, operator valued measures and bounded linear maps*, arXiv:1411.4559v2, Problem 2 on PDF page 19 [1]. The paper defines an operator-valued measure $E : \Sigma \rightarrow B(\mathcal{H})$ to be countably additive in the weak operator topology. Its question is:

Given a probability space (Ω, Σ, μ) and an operator-valued measure $E : (\Omega, \Sigma) \rightarrow B(\mathcal{H})$, does there exist an ultraweakly continuous map $\phi : L^\infty(\mu) \rightarrow B(\mathcal{H})$ such that $E(B) = \phi(\chi_B)$ for every $B \in \Sigma$? If not, determine necessary and sufficient conditions.

Let (Ω, Σ, μ) be a probability space and let $\phi : L^\infty(\mu) \rightarrow B(\mathcal{H})$ be ultraweakly continuous. Then it naturally induces an operator valued probability measure

$$E(B) = \phi(\chi_B), \quad \forall B \in \Sigma.$$

Problem 2. Let $E : (\Omega, \Sigma) \rightarrow B(\mathcal{H})$ be an operator valued measure. Is there an ultraweakly continuous map $\phi : L^\infty(\mu) \rightarrow B(\mathcal{H})$ that induces E on (Ω, Σ) ? If the answer is negative, then determine necessary and sufficient conditions for E to be induced by an ultraweakly continuous map?

Figure 1: Problem 2 in the source paper, PDF page 19.

2 Definitions

Write $E \ll \mu$ when

$$\mu(B) = 0 \implies E(B) = 0 \quad (B \in \Sigma).$$

We use the canonical preduals $L^1(\mu)$ of $L^\infty(\mu)$ and $S_1(\mathcal{H})$ of $B(\mathcal{H})$. Thus “ultraweakly continuous” means weak-star to weak-star continuous, or equivalently that the map is the adjoint of a bounded map $S_1(\mathcal{H}) \rightarrow L^1(\mu)$.

For a bounded simple function $s = \sum_j a_j \chi_{B_j}$ on a finite measurable partition, put

$$\int s dE := \sum_j a_j E(B_j).$$

The definition is independent of the chosen representation.

3 Proof intuition

The obstruction is forced by the quotient defining $L^\infty(\mu)$: the indicator of a null set is the zero element, so any inducing map must kill the operator assigned to that set. The striking point is that there is no further obstruction. Weak-operator countable additivity gives a uniform bound for all simple integrals. Absolute continuity then lets scalarizations by trace-class operators acquire L^1 Radon–Nikodym densities. Those densities form a bounded preadjoint, and its adjoint is the required normal map.

4 The characterization

Lemma 1 (Uniform semivariation). *Let $E : \Sigma \rightarrow B(\mathcal{H})$ be weak-operator countably additive. Then*

$$C_E := \sup \left\{ \left\| \int s dE \right\| : s \text{ simple, } |s| \leq 1 \right\} < \infty.$$

Consequently, for every $\rho \in S_1(\mathcal{H})$, the scalar set function

$$\nu_\rho(B) := \text{Tr}(E(B)\rho)$$

is a countably additive complex measure and

$$|\nu_\rho|(\Omega) \leq C_E \|\rho\|_1.$$

Proof. For $x \in \mathcal{H}$, the set function $E_x(B) = E(B)x$ is weakly countably additive. By the Orlicz–Pettis theorem it is a norm-countably-additive $\mathcal{H}$ -valued measure. Every countably additive Banach-space-valued measure has finite semivariation [2, Chapter I]; hence, for each fixed x ,

$$\sup_{s \text{ simple, } |s| \leq 1} \left\| \left(\int s dE \right) x \right\| < \infty.$$

The uniform boundedness principle applied to the family of simple-integration operators gives $C_E < \infty$.

Weak-operator countable additivity, together with uniform boundedness of the range of E , implies countable additivity after pairing with any trace-class operator: approximate that trace-class operator in trace norm by finite-rank operators. Thus ν_ρ is a complex measure. The standard partition formula for total variation and trace duality give

$$\begin{aligned} |\nu_\rho|(\Omega) &= \sup_{\substack{(B_j) \text{ finite partition} \\ |a_j| \leq 1}} \left| \sum_j a_j \text{Tr}(E(B_j)\rho) \right| \\ &= \sup_{s \text{ simple, } |s| \leq 1} \left| \text{Tr} \left(\left(\int s dE \right) \rho \right) \right| \\ &\leq C_E \|\rho\|_1. \end{aligned}$$

□

Theorem 1. *Let (Ω, Σ, μ) be a probability space, let $\mathcal{H}$ be any Hilbert space, and let $E : \Sigma \rightarrow B(\mathcal{H})$ be weak-operator countably additive. The following are equivalent.*

1. *There exists an ultraweakly continuous bounded linear map $\phi : L^\infty(\mu) \rightarrow B(\mathcal{H})$ satisfying $\phi(\chi_B) = E(B)$ for every $B \in \Sigma$.*
2. *$E \ll \mu$.*

When these conditions hold, ϕ is unique and is given by weak operator integration,

$$\langle \phi(f)x, y \rangle = \int_{\Omega} f d\langle E(\cdot)x, y \rangle.$$

If $E(\Omega) = I_{\mathcal{H}}$, then ϕ is unital.

Proof. Suppose first that such a ϕ exists. If $\mu(B) = 0$, then $\chi_B = 0$ as an element of $L^\infty(\mu)$, and therefore $E(B) = \phi(\chi_B) = 0$. Hence $E \ll \mu$.

Conversely assume $E \ll \mu$. For $\rho \in S_1(\mathcal{H})$, Lemma 1 gives a finite complex measure ν_ρ with $|\nu_\rho|(\Omega) \leq C_E \|\rho\|_1$. Absolute continuity of E implies $\nu_\rho \ll \mu$. By the scalar Radon–Nikodym theorem there is a unique $g_\rho \in L^1(\mu)$ such that

$$\nu_\rho(B) = \int_B g_\rho d\mu, \quad \|g_\rho\|_1 = |\nu_\rho|(\Omega) \leq C_E \|\rho\|_1.$$

Uniqueness of Radon–Nikodym derivatives shows that

$$\Psi : S_1(\mathcal{H}) \longrightarrow L^1(\mu), \quad \Psi(\rho) = g_\rho,$$

is linear, and the displayed estimate shows that it is bounded. Define

$$\phi := \Psi^* : L^\infty(\mu) \longrightarrow B(\mathcal{H})$$

using the canonical dualities. This map is bounded and ultraweakly continuous by construction. For $B \in \Sigma$ and $\rho \in S_1(\mathcal{H})$,

$$\begin{aligned} \text{Tr}(\phi(\chi_B)\rho) &= \int_B g_\rho d\mu \\ &= \nu_\rho(B) = \text{Tr}(E(B)\rho). \end{aligned}$$

Since trace-class operators separate points of $B(\mathcal{H})$, we obtain $\phi(\chi_B) = E(B)$.

The weak integration formula follows first for simple functions and then for all of L^∞ by uniform approximation by simple functions. The same approximation proves uniqueness. Finally, $\phi(1) = \phi(\chi_\Omega) = E(\Omega)$, proving the unital assertion. $\square$

5 Sharp negative example

Corollary 1. *The unrestricted existence question in Problem 2 has a negative answer, even when $\mathcal{H} = \mathbb{C}$ and E is a probability operator-valued measure.*

Proof. Take $\Omega = [0, 1]$ with Lebesgue measure and let

$$E(B) = \mathbf{1}_{\{0 \in B\}} I_{\mathcal{H}}.$$

This is a weak-operator countably additive probability measure, but $E(\{0\}) = I_{\mathcal{H}}$ while $\chi_{\{0\}} = 0$ in $L^\infty(\mu)$. No inducing map, normal or otherwise, can exist. $\square$

6 Verification and limitations

The proof was checked against the source definition of operator-valued measure and uses no separability assumption. The main verifier focus should be Lemma 1: the standard finite-semivariation theorem for countably additive vector measures, followed by uniform boundedness, supplies exactly the uniform constant required for the trace-class preadjoint.

A bounded exact-phrase and close-variant search was run on 22 July 2026 using the source title, “Problem 2”, “operator valued measure”, “ultraweakly continuous map”, and $L^\infty(\mu)$. It found the source and related later work on normal *dilations*, but no paper explicitly answering this induction problem. This packet therefore claims a full proof for the literal source question, not priority or publication-level novelty. The characterization is a short consequence of classical vector-measure theory and may be folklore.

References

- [1] D. Han, D. R. Larson, B. Liu, and R. Liu, *Dilations of frames, operator valued measures and bounded linear maps*, arXiv:1411.4559v2 (2014), Problem 2, p. 19.
- [2] J. Diestel and J. J. Uhl, Jr., *Vector Measures*, Mathematical Surveys, vol. 15, American Mathematical Society, 1977.